

Advanced quadratic equations

№1. Solve the following equations:

1. $(x + 13)^2 + (x + 13) = 12;$

3. $-x^4 + 125x^2 - 2500 = 0;$

2. $x^4 - 10x^2 + 9 = 0;$

4. $14(x - 11)^4 + x^2 + 103 = 22x;$

№2. Solve an equation $f(2f(x)) = 117$ for a function $f(x) = x^2 + 2x - 3$.

2:4-

№3. Solve an equation $(x + 2)^4 + x^4 = 82$.

1:8-